

Volatility Forecasting with GARCH Models: In-Sample Selection versus Out-of-Sample Evaluation under Robust Loss Functions

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Abstract

This study compares twelve GARCH-family specifications: three conditional-variance models, GARCH, EGARCH, and APARCH, together with a GARCH-in-mean extension, each estimated under Gaussian, Student- t , and GED innovations. Each specification is estimated separately for two assets of deliberately different nature: Goldman Sachs (GS), an equity exposed to financial-sector shocks, and the iShares Silver Trust (SLV), a commodity-linked ETF. Models are estimated on 2020–2024 data and evaluated on one-step-ahead rolling volatility forecasts over 2025–2026 using loss functions that are robust to the squared-return proxy in the sense of Patton (2011): QLIKE as the primary criterion and MSE as a robustness check. Two results stand out. For GS, asymmetric Student- t models dominate both in-sample information criteria and out-of-sample QLIKE comparisons, despite exhibiting the weakest residual diagnostics. For SLV, an apparent out-of-sample superiority of asymmetric models, obtained under the non-robust MAE loss, vanishes entirely under both robust losses: the finding was an artifact of the evaluation criterion, not a property of the forecasts. The choice of loss function thus changes the study’s conclusion for one of the two assets, providing a direct empirical illustration of why robustness to imperfect volatility proxies matters in applied forecast comparison.

1 Introduction

Conditional volatility is unobservable, which makes volatility forecast evaluation a comparison against a proxy rather than against the target itself. Patton (2011) shows that this technical point has sharp practical consequences: unless the loss function belongs to a restricted robust class, the ranking of two forecasts under a noisy proxy (such as the daily squared return) can differ systematically from their ranking under the true conditional variance. Among commonly used criteria, MSE and QLIKE are robust, while MAE is not.

This paper takes that result seriously in a standard applied setting. We estimate a grid of twelve GARCH-family specifications on two assets with deliberately different risk profiles: Goldman Sachs Group Inc. (GS) and the iShares Silver Trust (SLV), and evaluate their one-step-ahead volatility forecasts over an out-of-sample period that happens to contain an extreme event: a silver price collapse in early 2026 with a daily return near -30% . This configuration turns out to be an ideal setting for Patton’s argument. Under the non-robust MAE loss, asymmetric models appear significantly superior for SLV. Under QLIKE and MSE, the superiority disappears.

A second theme concerns the relationship between in-sample selection and out-of-sample performance. For GS, information criteria and residual diagnostics point in mutually inconsistent directions: the specifications preferred by AIC/BIC are precisely those with the worst residual behaviour. We argue that in such cases out-of-sample evaluation under a robust loss is the natural arbiter, and for GS it delivers a clear answer in favour of the asymmetric Student- t class.

The remainder of the paper is organised as follows. Section 2 describes the data and the train/test design, presents the model grid, the forecasting scheme, and the evaluation methodology.

Section 3 reports in-sample results, Section 4 the out-of-sample comparison, and Section 5 concludes.

2 Data and evaluation design

We use daily closing prices for GS and SLV from January 2, 2020 to February 27, 2026 (1,547 observations per asset), transformed into percentage log-returns. The sample is split into a training set through December 31, 2024 (1,257 returns) and a test set covering January 2025 – February 2026 (289 returns). All estimation, specification choices, and model selection use the training set only, while the test set is reserved for forecast evaluation.

The split is not a formality. Full-sample skewness for SLV is approximately -2.4 , driven entirely by the extreme returns of early 2026, while training-sample skewness is a moderate -0.34 . Selecting the innovation distribution on the full sample would have imported information unavailable at estimation time, resulting in a look-ahead bias that would distort both model selection and the subsequent forecast comparison. Both series display the stylized facts of daily financial returns: means indistinguishable from zero, pronounced excess kurtosis (approximately 9.9 for GS and 5.4 for SLV in the training sample), rejection of normality by the Jarque–Bera test, and strong volatility clustering, with Ljung–Box tests on squared returns and Engle’s LM-ARCH tests rejecting at all considered lags. Descriptive statistics are reported in Table 1.

	GS	SLV
n	1257	1257
Mean	0.071	0.036
Median	0.029	0.039
Std. dev.	2.084	2.013
Skewness	-0.035	-0.337
Excess kurtosis	9.85	5.42
Min	-13.59	-14.61
Max	16.20	8.75

Table 1: Descriptive statistics of training-sample percentage log-returns (Jan 2020–Dec 2024). Kurtosis is reported in excess of 3.

2.1 Model grid

For each asset, we estimate twelve models obtained by combining three conditional-variance dynamics: GARCH, EGARCH, APARCH, plus a GARCH-in-mean extension of the standard GARCH model with three innovation distributions. Except for the GARCH-in-mean specification, the conditional mean is constant and given by

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t,$$

where $z_t \sim \text{IID}(0, 1)$. The model equations below are expressed using the parameterization adopted by the `rugarch` package of Galanos (2026) used for estimation.

The symmetric GARCH(1,1) of Bollerslev (1986):

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2. \quad (1)$$

The GARCH-in-mean model of Engle, Lilien and Robins (1987), which retains the GARCH(1,1) variance equation and adds a risk-premium term to the conditional mean,

$$r_t = \mu + \lambda \sigma_t + \varepsilon_t, \quad (2)$$

so that $\lambda > 0$ corresponds to compensation for conditional risk.

The EGARCH model of Nelson (1991):

$$\log \sigma_t^2 = \omega + \alpha_1 z_{t-1} + \gamma_1 (|z_{t-1}| - \mathbb{E}|z_{t-1}|) + \beta_1 \log \sigma_{t-1}^2, \quad (3)$$

where α_1 captures the sign effect, $\alpha_1 < 0$ implies that negative shocks raise volatility by more than positive shocks of equal magnitude, and γ_1 the magnitude effect.

The APARCH model of Ding, Granger and Engle (1993):

$$\sigma_t^\delta = \omega + \alpha_1 (|\varepsilon_{t-1}| - \gamma_1 \varepsilon_{t-1})^\delta + \beta_1 \sigma_{t-1}^\delta, \quad \delta > 0, \gamma_1 \in (-1, 1), \quad (4)$$

in which $\gamma_1 > 0$ implies a stronger volatility response to negative shocks, and which includes the symmetric GARCH(1,1) at $\delta = 2$, $\gamma_1 = 0$.

Innovations z_t are Gaussian, Student- t , or generalized error distributed (GED). Given the moderate training-sample skewness of both assets, asymmetric innovation distributions are excluded on parsimony grounds.

2.2 Forecasting scheme

One-step-ahead volatility forecasts for the test period are produced by a rolling procedure with an expanding estimation window: specifications are fixed at their training-sample form, while parameters are re-estimated every five observations as new data enter the window. This mimics real-time use, at each forecast date, only information available at that date is used, while keeping the computational cost of 12 models $\times$ 2 assets $\times$ 58 re-estimations manageable.

2.3 Loss functions and forecast comparison

Let $h_t = \hat{\sigma}_t^2$ denote a one-step-ahead conditional variance forecast and r_t^2 the squared-return proxy of the latent variance. Following Patton (2011), we restrict attention to loss functions that are *robust* to proxy noise, i.e. whose expected-loss ranking of any two forecasts coincides with the ranking under the true conditional variance. Our primary criterion is

$$\text{QLIKE:} \quad L(h_t, r_t^2) = \log h_t + \frac{r_t^2}{h_t}, \quad (5)$$

with

$$\text{MSE:} \quad L(h_t, r_t^2) = (h_t - r_t^2)^2 \quad (6)$$

as a robustness check. Two considerations motivate this hierarchy. First, the MSE loss depends on the forecast error $h_t - r_t^2$, whose variance is proportional to σ_t^4 . Sample-average MSE comparisons are therefore dominated by the most volatile observations, a serious concern in our test sample, where a single SLV return near -30% accounts for the bulk of total squared-error loss. QLIKE depends on the standardised error r_t^2/h_t , whose conditional variance is approximately constant across volatility regimes, so that extreme episodes receive weight without monopolizing the comparison. Second, MAE is *not* robust: its optimal forecast is the conditional median of r_t^2 , which for fat-tailed returns lies far below the conditional variance (approximately $0.45 \sigma_t^2$ under conditional normality). MAE therefore systematically rewards models that underpredict variance. As Section 4 shows, this property alone created an apparent model ranking for SLV. To make the contrast visible, all Diebold–Mariano comparisons are therefore reported under MAE as well (Tables 9 and 12), alongside the two robust criteria.

Pairwise predictive-accuracy comparisons use the Diebold and Mariano (1995) test applied directly to loss differentials $d_t = L_{i,t} - L_{j,t}$, with the long-run variance estimated by the Newey and West (1987) estimator (lag truncation $\lfloor T^{1/3} \rfloor$). Tests are one-sided and the entry (i, j) of each reported matrix is the p -value against the alternative that model i has strictly larger

expected loss than model j . With 66 distinct pairs per asset and loss, the matrices are reported descriptively: conclusions are drawn only for model *blocks* specified in advance (symmetric versus asymmetric variance equations, Gaussian versus heavy-tailed innovations) and supported by internally consistent sets of pairwise results.

3 In-sample results

3.1 Goldman Sachs

Estimation results for GS deliver a coherent message on the innovation distribution and a conflicted one on the variance equation. The Student- t shape parameter is estimated around 5.9, indicating substantial tail thickness, consistent with the pronounced excess kurtosis of the series. QQ-plots of standardized residuals show that the t resolves the tail deviations that the Gaussian leaves and that the GED only partially absorbs. Across all three information criteria the asymmetric t models dominate: APARCH+ t and EGARCH+ t occupy the top two ranks, followed by the symmetric t models, with all Gaussian specifications ranked last (Table 2).

The asymmetry evidence is significant. In the EGARCH+ t specification the sign parameter is estimated at $\hat{\alpha}_1 = -0.088$ (p-value < 0.001), indicating that negative shocks raise volatility by more than positive shocks of equal size.

The conflict emerges from residual diagnostics (Table 3). The strongest and most systematic diagnostic failures are concentrated in the specifications preferred by the information criteria, namely EGARCH and APARCH with Student- t or GED innovations. For these models, the LM-ARCH test rejects at short lags, while the positive sign-bias and joint sign-bias tests also indicate remaining residual asymmetry. This is particularly noteworthy because these specifications are explicitly designed to capture asymmetric volatility responses. The pattern is consistent with a reallocation mechanism: a heavy-tailed likelihood can rationalise extreme observations as ordinary draws from a fat tail rather than as episodes of elevated conditional variance, so the variance equation reacts less, and structure that the dynamics should absorb, including part of the asymmetry, remains in the standardized residuals. Gaussian specifications display the mirror image: clean residual diagnostics achieved by the variance dynamics, at the cost of a clearly misspecified distribution.

In-sample model selection for GS is therefore ill-posed: likelihood-based criteria and specification tests disagree, and they disagree because they measure different things. This motivates treating out-of-sample forecast performance under a robust loss as the arbiter.

	AIC	BIC	HQ
sGARCH+N	11	10	11
GARCH-M+N	12	12	12
eGARCH+N	10	11	10
apARCH+N	9	9	9
sGARCH+t	4	3	3
GARCH-M+t	3	4	4
eGARCH+t	2	1	2
apARCH+t	1	2	1
sGARCH+GED	8	7	7
GARCH-M+GED	7	8	8
eGARCH+GED	6	5	6
apARCH+GED	5	6	5

Table 2: Information criteria ranks (AIC, BIC, Hannan–Quinn), GS training sample.

	persistence	unc_sd	ARCH_p_1	ARCH_p_5	ARCH_p_22	LB2_p_5	LB2_p_10	sign_bias_p	neg_bias_p	pos_bias_p	joint_p
sGARCH+N	0.8931	1.9670	0.5414	0.6166	0.6641	0.6028	0.7460	0.8093	0.1897	0.4710	0.4427
GARCH-M+N	0.9011	1.9657	0.4526	0.6247	0.6998	0.6091	0.7336	0.8507	0.2006	0.3564	0.4490
eGARCH+N	0.9382	1.9863	0.1455	0.6110	0.8911	0.6064	0.8441	0.9677	0.1541	0.0447**	0.1004
apARCH+N	0.9269	2.0045	0.2137	0.5858	0.8131	0.5749	0.7855	1.0000	0.2572	0.0984	0.2419
sGARCH+t	0.8913	1.8875	0.2338	0.7440	0.8045	0.7364	0.8718	0.8246	0.0428**	0.1775	0.0949
GARCH-M+t	0.9035	1.8819	0.1722	0.7250	0.8408	0.7198	0.8613	0.8031	0.0468**	0.1017	0.0782
eGARCH+t	0.9694	1.7960	0.0271**	0.3765	0.9047	0.3859	0.7578	0.9712	0.0855	0.0053**	0.0093**
apARCH+t	0.9591	1.8401	0.0410**	0.4433	0.8908	0.4546	0.7858	0.9331	0.1279	0.0087**	0.0176**
sGARCH+GED	0.8894	1.8882	0.3407	0.7224	0.7728	0.7098	0.8386	0.8738	0.0773	0.3014	0.2037
GARCH-M+GED	0.9022	1.8824	0.2535	0.7221	0.8234	0.7116	0.8359	0.9336	0.0622	0.2330	0.1755
eGARCH+GED	0.9631	1.8076	0.0421**	0.4559	0.9121	0.4642	0.7998	0.9927	0.0914	0.0088**	0.0164**
apARCH+GED	0.9463	1.8474	0.0815	0.5442	0.8820	0.5481	0.8151	0.9838	0.1696	0.0197**	0.0496**

Table 3: Residual diagnostics, GS training sample: LM-ARCH and Ljung–Box p -values on squared standardized residuals, sign-bias tests. Asterisks flag $p < 0.05$.

3.2 iShares Silver Trust

For SLV the in-sample evidence is internally consistent and points the opposite way. Asymmetry parameters are statistically indistinguishable from zero, and the information criteria reward parsimony: the symmetric GARCH(1,1)+ t ranks first on all three criteria, with GARCH-M close behind (archm parameter not significantly different from 0) and the asymmetric specifications penalized for parameters the data do not support. On the distributional margin the evidence is mixed: QQ-plots mildly favour the GED, consistent with SLV’s more moderate kurtosis, while the likelihood-based criteria prefer the t .

Residual diagnostics (Table 5) again concentrate their failures in the asymmetric heavy-tailed specifications: EGARCH+ t , EGARCH+GED and APARCH+ t reject LM-ARCH and Ljung–Box tests across essentially all lags, while the symmetric t /GED models show at most a long-lag rejection (for 22 lags) and Gaussian models pass throughout. For SLV, in-sample evidence coherently selects the symmetric GARCH(1,1)+ t . The persistence is high ($\hat{\alpha} + \hat{\beta} \approx 0.96$), consistent with slowly decaying volatility shocks in the commodity ETF.

	AIC	BIC	HQ
sGARCH+N	9	9	9
GARCH-M+N	10	10	10
eGARCH+N	12	12	12
apARCH+N	11	11	11
sGARCH+t	1	1	1
GARCH-M+t	2	3	2
eGARCH+t	4	4	4
apARCH+t	3	6	5
sGARCH+GED	5	2	3
GARCH-M+GED	6	5	6
eGARCH+GED	8	7	8
apARCH+GED	7	8	7

Table 4: Information criteria ranks (AIC, BIC, Hannan–Quinn), SLV training sample.

	persistence	unc_sd	ARCH_p_1	ARCH_p_5	ARCH_p_22	LB2_p_5	LB2_p_10	sign_bias_p	neg_bias_p	pos_bias_p	joint_p
sGARCH+N	0.9697	1.9999	0.4916	0.6905	0.1079	0.6826	0.8799	0.4307	0.5990	0.2153	0.6065
GARCH-M+N	0.9706	2.0030	0.4776	0.6688	0.1040	0.6606	0.8607	0.5290	0.6355	0.2460	0.6470
eGARCH+N	0.9782	2.3267	0.2455	0.2000	0.0587	0.1842	0.4855	0.4963	0.9965	0.3161	0.7875
apARCH+N	0.9678	1.9783	0.3912	0.5226	0.1401	0.5097	0.7765	0.4985	0.8324	0.3161	0.7875
sGARCH+t	0.9637	2.0345	0.1216	0.0986	0.0037**	0.0723	0.3055	0.5363	0.8567	0.0570	0.2816
GARCH-M+t	0.9643	2.0348	0.1146	0.0843	0.0035**	0.0617	0.2797	0.5337	0.8636	0.0592	0.2889
eGARCH+t	0.9789	1.9752	0.0030**	0.0000**	0.0000**	0.0000**	0.0001**	0.7708	0.1537	0.1027	0.1912
apARCH+t	0.9686	2.0165	0.0161**	0.0031**	0.0015**	0.0014**	0.0153**	0.7291	0.3083	0.1245	0.3330
sGARCH+GED	0.9584	1.9311	0.3050	0.3585	0.0287**	0.3355	0.7017	0.4786	0.8367	0.1165	0.4583
GARCH-M+GED	0.9589	1.9314	0.2949	0.3238	0.0278**	0.3033	0.6636	0.5299	0.8497	0.1314	0.4858
eGARCH+GED	0.9754	1.8973	0.0430**	0.0076**	0.0033**	0.0044**	0.0390**	0.6215	0.4715	0.1775	0.4966
apARCH+GED	0.9583	1.9303	0.1703	0.1461	0.0421**	0.1291	0.4229	0.5908	0.7835	0.2142	0.6545

Table 5: Residual diagnostics, SLV training sample: LM-ARCH and Ljung–Box p -values on squared standardized residuals, and sign-bias tests. Asterisks flag $p < 0.05$.

4 Out-of-sample results

4.1 Goldman Sachs: robust losses sharpen the in-sample verdict

Table 6 reports mean QLIKE and MSE over the 289 forecast days. For GS the point ranking is unambiguous under both losses: EGARCH+ t attains the lowest mean QLIKE (2.239) and the lowest MSE, with the asymmetric block (EGARCH and APARCH, t and GED) filling the top ranks and all symmetric specifications clustered behind (mean QLIKE 2.286–2.288).

Table 6: Mean out-of-sample loss by model, 2025–2026 test period (289 one-step-ahead forecasts).

Model	GS			SLV		
	QLIKE	MSE	MAE	QLIKE	MSE	MAE
sGARCH+N	2.2880	119.42	4.626	2.8637	4771.5	12.701
GARCH-M+N	2.2882	119.34	4.634	2.8796	4772.8	12.656
eGARCH+N	2.2548	116.76	4.439	2.7840	4687.9	11.743
apARCH+N	2.2641	117.95	4.460	2.8035	4691.3	11.610
sGARCH+ t	2.2865	118.86	4.518	2.8972	4718.9	11.855
GARCH-M+ t	2.2866	118.82	4.529	2.9222	4728.1	11.870
eGARCH+ t	2.2390	116.73	4.364	2.7785	4674.1	11.085
apARCH+ t	2.2481	117.39	4.371	2.7679	4664.0	11.159
sGARCH+GED	2.2881	118.94	4.532	2.8941	4730.6	12.009
GARCH-M+GED	2.2882	118.85	4.534	2.9168	4738.9	12.027
eGARCH+GED	2.2441	116.97	4.368	2.7867	4680.0	11.158
apARCH+GED	2.2532	117.38	4.394	2.8009	4683.7	11.178

The Diebold-Mariano matrices (Tables 7 and 8) show how the two robust losses relate: they agree in direction and differ in power. Under QLIKE, every symmetric specification is significantly outperformed by EGARCH+ t (p -values between 0.037 and 0.045), with APARCH+ t marginal ($p \approx 0.05$ –0.07) and the asymmetric models mutually indistinguishable. Under MSE the same comparisons point the same way but lose significance ($p \approx 0.12$ –0.19). The GS conclusion therefore both survives the move to robust losses and sharpens: the asymmetric- t advantage is detectable under the high-power robust loss and directionally confirmed under the low-power one.

The result is also strong in a specific sense. QLIKE is an asymmetric criterion that penalises *under*-prediction of variance most heavily, exactly the failure mode of asymmetric models around

large positive shocks, where their attenuated response leaves forecasts below realised volatility (Figure 1). EGARCH+ t wins the QLIKE comparison despite this handicap.

	sGARCH+N	GARCH-M+N	eGARCH+N	apARCH+N	sGARCH+t	GARCH-M+t	eGARCH+t	apARCH+t	sGARCH+GED	GARCH-M+GED	eGARCH+GED	apARCH+GED
sGARCH+N	NA	0.548	0.056	0.088	0.417	0.429	0.039	0.06	0.501	0.511	0.043	0.066
GARCH-M+N	0.452	NA	0.054	0.084	0.413	0.422	0.037	0.058	0.487	0.498	0.041	0.064
eGARCH+N	0.944	0.946	NA	0.927	0.934	0.941	0.074	0.208	0.941	0.948	0.081	0.394
apARCH+N	0.912	0.916	0.073	NA	0.902	0.911	0.053	0.073	0.916	0.925	0.05	0.069
sGARCH+t	0.583	0.587	0.066	0.098	NA	0.527	0.045	0.065	0.697	0.692	0.048	0.07
GARCH-M+t	0.571	0.578	0.059	0.089	0.473	NA	0.041	0.059	0.645	0.7	0.043	0.063
eGARCH+t	0.961	0.963	0.926	0.947	0.955	0.959	NA	0.919	0.958	0.963	0.915	0.93
apARCH+t	0.94	0.942	0.792	0.927	0.935	0.941	0.081	NA	0.94	0.946	0.16	0.898
sGARCH+GED	0.499	0.513	0.059	0.084	0.303	0.355	0.042	0.06	NA	0.528	0.044	0.064
GARCH-M+GED	0.489	0.502	0.052	0.075	0.308	0.3	0.037	0.054	0.472	NA	0.039	0.057
eGARCH+GED	0.957	0.959	0.919	0.95	0.952	0.957	0.085	0.84	0.956	0.961	NA	0.927
apARCH+GED	0.934	0.936	0.606	0.931	0.93	0.937	0.07	0.102	0.936	0.943	0.073	NA

Table 7: Diebold–Mariano p -values, GS, QLIKE loss. Entry (i, j) tests the alternative that model i (row) has larger expected loss than model j (column).

	sGARCH+N	GARCH-M+N	eGARCH+N	apARCH+N	sGARCH+t	GARCH-M+t	eGARCH+t	apARCH+t	sGARCH+GED	GARCH-M+GED	eGARCH+GED	apARCH+GED
sGARCH+N	NA	0.43	0.16	0.276	0.349	0.352	0.194	0.261	0.319	0.32	0.21	0.248
GARCH-M+N	0.57	NA	0.161	0.283	0.377	0.367	0.193	0.264	0.363	0.329	0.212	0.249
eGARCH+N	0.84	0.839	NA	0.949	0.913	0.93	0.479	0.768	0.886	0.897	0.639	0.815
apARCH+N	0.724	0.717	0.051	NA	0.776	0.795	0.122	0.251	0.74	0.738	0.112	0.199
sGARCH+t	0.651	0.623	0.087	0.224	NA	0.465	0.141	0.222	0.575	0.491	0.148	0.2
GARCH-M+t	0.648	0.633	0.07	0.205	0.535	NA	0.116	0.201	0.572	0.523	0.126	0.173
eGARCH+t	0.806	0.807	0.521	0.878	0.859	0.884	NA	0.892	0.835	0.853	0.767	0.899
apARCH+t	0.739	0.736	0.232	0.749	0.778	0.799	0.108	NA	0.753	0.762	0.115	0.488
sGARCH+GED	0.681	0.637	0.114	0.26	0.425	0.428	0.165	0.247	NA	0.432	0.178	0.228
GARCH-M+GED	0.68	0.671	0.103	0.262	0.509	0.477	0.147	0.238	0.568	NA	0.166	0.215
eGARCH+GED	0.79	0.788	0.361	0.888	0.852	0.874	0.233	0.885	0.822	0.834	NA	0.902
apARCH+GED	0.752	0.751	0.185	0.801	0.8	0.827	0.101	0.512	0.772	0.785	0.098	NA

Table 8: Diebold–Mariano p -values, GS, MSE loss. Entry (i, j) tests the alternative that model i (row) has larger expected loss than model j (column).

	sGARCH+N	GARCH-M+N	eGARCH+N	apARCH+N	sGARCH+t	GARCH-M+t	eGARCH+t	apARCH+t	sGARCH+GED	GARCH-M+GED	eGARCH+GED	apARCH+GED
sGARCH+N	NA	0.798	0.01	0.008	0.031	0.028	0.003	0.003	0.011	0.016	0.003	0.003
GARCH-M+N	0.202	NA	0.009	0.007	0.032	0.026	0.003	0.003	0.014	0.014	0.003	0.002
eGARCH+N	0.99	0.991	NA	0.87	0.933	0.953	0.017	0.013	0.951	0.951	0.005	0.027
apARCH+N	0.992	0.993	0.13	NA	0.872	0.919	0.014	0.006	0.924	0.929	0.007	0.004
sGARCH+t	0.969	0.968	0.067	0.128	NA	0.772	0.015	0.017	0.799	0.79	0.011	0.022
GARCH-M+t	0.972	0.974	0.047	0.081	0.228	NA	0.008	0.009	0.585	0.694	0.006	0.01
eGARCH+t	0.997	0.997	0.983	0.986	0.985	0.992	NA	0.712	0.987	0.99	0.677	0.916
apARCH+t	0.997	0.997	0.987	0.994	0.983	0.991	0.288	NA	0.986	0.989	0.39	0.969
sGARCH+GED	0.989	0.986	0.049	0.076	0.201	0.415	0.013	0.014	NA	0.568	0.01	0.015
GARCH-M+GED	0.984	0.986	0.049	0.071	0.21	0.306	0.01	0.011	0.432	NA	0.008	0.012
eGARCH+GED	0.997	0.997	0.995	0.993	0.989	0.994	0.323	0.61	0.99	0.992	NA	0.933
apARCH+GED	0.997	0.998	0.973	0.996	0.978	0.99	0.084	0.031	0.985	0.988	0.067	NA

Table 9: Diebold–Mariano p -values, GS, MAE loss. Entry (i, j) tests the alternative that model i (row) has larger expected loss than model j (column).

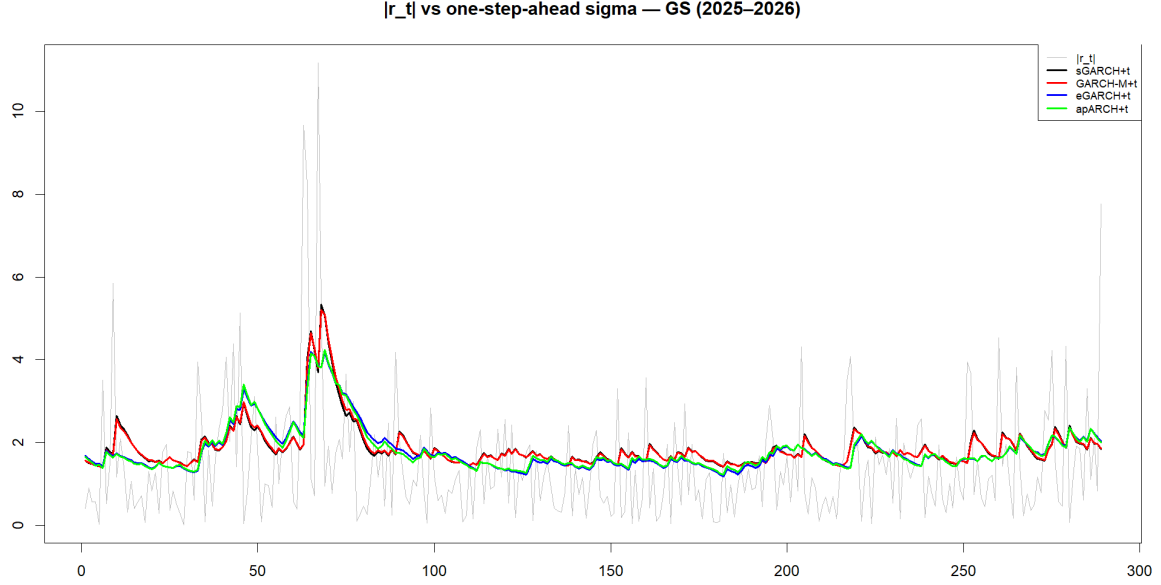


Figure 1: Absolute returns and one-step-ahead volatility forecasts, GS, 2025–2026, Student- t specifications. Symmetric models react more strongly to the extreme positive return of April 2025; asymmetric models attenuate the response, consistent with their estimated asymmetry effect.

4.2 SLV: a finding that depends on the loss function

The SLV comparison is where the choice of evaluation criterion becomes decisive. Under the MAE loss, the asymmetric specifications with the GED distribution appear superior to all symmetric ones out-of-sample, despite in-sample asymmetry parameters indistinguishable from zero, suggesting a divergence between in-sample selection and out-of-sample performance that would have been undetectable in real time.

Under the robust losses the problem disappears. Mean losses still rank the asymmetric models first (Table 6), but no pairwise Diebold–Mariano comparison is significant at the 5% level under either QLIKE or MSE (Tables 10–11). The MAE result was an artifact of the loss function. Read jointly with the GS results, the pattern is coherent: under MAE both assets display strongly significant asymmetric-model superiority. Under the robust losses, the effect survives at attenuated strength where in-sample evidence supports it (GS) and vanishes where it does not (SLV).

The mechanism is precisely Patton’s: MAE’s optimal forecast is the conditional *median* of r_t^2 , far below the conditional variance for fat-tailed returns, so MAE rewards downward-biased volatility forecasts, and the asymmetric heavy-tailed models, whose smoother variance paths systematically under-react (Section 3), were being rewarded for their bias rather than their accuracy.

The corrected conclusion also rehabilitates in-sample selection for this asset: the training data pointed to the symmetric GARCH(1,1)+ t , and under robust evaluation no competitor significantly improves on it. The apparent real-time selection failure was created by the evaluation criterion, not by the models.

Finally, the extreme event of early 2026, a return near -30% , double the largest absolute return in the training sample, dominates the level of all loss measures (mean MSE for SLV is roughly 40 times that of GS) and illustrates a structural limit shared by the entire model class: GARCH models are univariate conditional-volatility models calibrated on history and cannot anticipate breaks in the data-generating process. The episode also reveals a purely mechanical difference in post-shock behaviour (Figure 2): the symmetric GARCH variance responds through

$\alpha \varepsilon_{t-1}^2$, quadratic in the shock, and jumps violently after the crash, while EGARCH moves log-variance by terms linear in the standardised shock and adjusts far more slowly. Under QLIKE the ex-post over-reaction of the symmetric models is cheap and the slowness of the asymmetric ones expensive.

	sGARCH+N	GARCH-M+N	eGARCH+N	apARCH+N	sGARCH+t	GARCH-M+t	eGARCH+t	apARCH+t	sGARCH+GED	GARCH-M+GED	eGARCH+GED	apARCH+GED
sGARCH+N	NA	0.857	0.097	0.105	0.914	0.902	0.131	0.137	0.882	0.88	0.103	0.123
GARCH-M+N	0.143	NA	0.103	0.112	0.888	0.913	0.131	0.138	0.876	0.888	0.108	0.126
eGARCH+N	0.903	0.897	NA	0.837	0.912	0.906	0.395	0.317	0.9	0.896	0.62	0.792
apARCH+N	0.895	0.888	0.163	NA	0.911	0.902	0.213	0.19	0.894	0.89	0.189	0.386
sGARCH+t	0.086	0.112	0.088	0.089	NA	0.866	0.109	0.115	0.336	0.787	0.089	0.098
GARCH-M+t	0.098	0.087	0.094	0.098	0.134	NA	0.112	0.118	0.081	0.203	0.096	0.105
eGARCH+t	0.869	0.869	0.605	0.787	0.891	0.888	NA	0.276	0.877	0.877	0.686	0.806
apARCH+t	0.863	0.862	0.683	0.81	0.885	0.882	0.724	NA	0.871	0.871	0.723	0.833
sGARCH+GED	0.118	0.124	0.1	0.106	0.664	0.919	0.123	0.129	NA	0.87	0.103	0.115
GARCH-M+GED	0.12	0.112	0.104	0.11	0.213	0.797	0.123	0.129	0.13	NA	0.107	0.118
eGARCH+GED	0.897	0.892	0.38	0.811	0.911	0.904	0.314	0.277	0.897	0.893	NA	0.8
apARCH+GED	0.877	0.874	0.208	0.614	0.902	0.895	0.194	0.167	0.885	0.882	0.2	NA

Table 10: Diebold–Mariano p -values, SLV, QLIKE loss. Entry (i, j) tests the alternative that model i (row) has larger expected loss than model j (column).

	sGARCH+N	GARCH-M+N	eGARCH+N	apARCH+N	sGARCH+t	GARCH-M+t	eGARCH+t	apARCH+t	sGARCH+GED	GARCH-M+GED	eGARCH+GED	apARCH+GED
sGARCH+N	NA	0.579	0.068	0.068	0.081	0.122	0.104	0.096	0.087	0.139	0.089	0.09
GARCH-M+N	0.421	NA	0.075	0.074	0.068	0.1	0.108	0.102	0.066	0.105	0.092	0.093
eGARCH+N	0.932	0.925	NA	0.65	0.771	0.796	0.339	0.232	0.826	0.839	0.376	0.429
apARCH+N	0.932	0.926	0.35	NA	0.784	0.811	0.285	0.203	0.838	0.853	0.286	0.335
sGARCH+t	0.919	0.932	0.229	0.216	NA	0.887	0.216	0.196	0.917	0.941	0.196	0.202
GARCH-M+t	0.878	0.9	0.204	0.189	0.113	NA	0.192	0.18	0.61	0.901	0.171	0.175
eGARCH+t	0.896	0.892	0.661	0.715	0.784	0.808	NA	0.202	0.816	0.832	0.686	0.733
apARCH+t	0.904	0.898	0.768	0.797	0.804	0.82	0.798	NA	0.83	0.842	0.786	0.803
sGARCH+GED	0.913	0.934	0.174	0.162	0.083	0.39	0.184	0.17	NA	0.901	0.163	0.166
GARCH-M+GED	0.861	0.895	0.161	0.147	0.059	0.099	0.168	0.158	0.099	NA	0.147	0.149
eGARCH+GED	0.911	0.908	0.624	0.714	0.804	0.829	0.314	0.214	0.837	0.853	NA	0.865
apARCH+GED	0.91	0.907	0.571	0.665	0.798	0.825	0.267	0.197	0.834	0.851	0.135	NA

Table 11: Diebold–Mariano p -values, SLV, MSE loss. Entry (i, j) tests the alternative that model i (row) has larger expected loss than model j (column).

	sGARCH+N	GARCH-M+N	eGARCH+N	apARCH+N	sGARCH+t	GARCH-M+t	eGARCH+t	apARCH+t	sGARCH+GED	GARCH-M+GED	eGARCH+GED	apARCH+GED
sGARCH+N	NA	0.057	0.056	0.046	0.048	0.048	0.045	0.046	0.034	0.035	0.038	0.037
GARCH-M+N	0.943	NA	0.059	0.047	0.049	0.05	0.045	0.047	0.034	0.035	0.039	0.038
eGARCH+N	0.944	0.941	NA	0.018	0.692	0.706	0.05	0.048	0.811	0.821	0.028	0.025
apARCH+N	0.954	0.953	0.982	NA	0.863	0.866	0.064	0.064	0.893	0.898	0.036	0.03
sGARCH+t	0.952	0.951	0.308	0.137	NA	0.806	0.046	0.051	0.88	0.89	0.035	0.034
GARCH-M+t	0.952	0.95	0.294	0.134	0.194	NA	0.047	0.052	0.869	0.884	0.036	0.035
eGARCH+t	0.955	0.955	0.95	0.936	0.954	0.953	NA	0.894	0.943	0.943	0.772	0.78
apARCH+t	0.954	0.953	0.952	0.936	0.949	0.948	0.106	NA	0.938	0.939	0.491	0.602
sGARCH+GED	0.966	0.966	0.189	0.107	0.12	0.131	0.057	0.062	NA	0.905	0.048	0.047
GARCH-M+GED	0.965	0.965	0.179	0.102	0.11	0.116	0.057	0.061	0.095	NA	0.047	0.046
eGARCH+GED	0.962	0.961	0.972	0.964	0.965	0.964	0.228	0.509	0.952	0.953	NA	0.757
apARCH+GED	0.963	0.962	0.975	0.97	0.966	0.965	0.22	0.398	0.953	0.954	0.243	NA

Table 12: Diebold–Mariano p -values, SLV, MAE loss. Entry (i, j) tests the alternative that model i (row) has larger expected loss than model j (column).

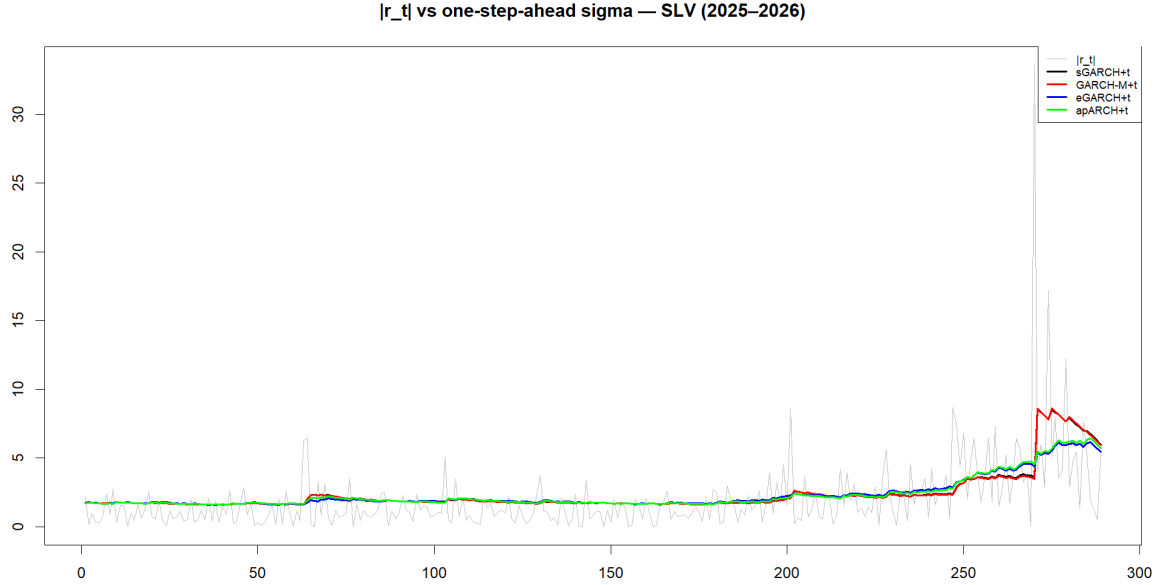


Figure 2: Absolute returns and one-step-ahead volatility forecasts, SLV, 2025–2026, Student- t specifications.

5 Conclusion

Three lessons emerge. First, in-sample selection criteria can conflict: for GS, the specifications favoured by information criteria are exactly those with the weakest residual diagnostics, a tension we trace to the tail flexibility of heavy-tailed likelihoods. When in-sample criteria have conflicting results, out-of-sample evaluation under a robust loss provides the natural resolution, and for GS it does so clearly, in favour of EGARCH with Student- t innovations.

Second, and centrally, the evaluation loss is not a neutral implementation detail. Replacing MAE with the proxy-robust QLIKE and MSE criteria of Patton (2011) reversed this study’s conclusion for one of its two assets: the apparently significant out-of-sample superiority of asymmetric models for SLV, and with it an apparent failure of in-sample selection, disappeared entirely. Empirical rankings of volatility forecasts obtained under non-robust losses should be treated as unreliable by default.

Third, the extreme SLV event of 2026 delineates what forecast evaluation can and cannot say: within-regime accuracy can be measured and compared while robustness to structural breaks cannot be inferred from any of it, since all univariate conditional-volatility models fail such episodes together.

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